

Real-time Detailed Video Analysis of Fruit Flies: a Reproduction

UE24CS352A Machine Learning mini-project · [Team member 1, SRN] · [Team member 2, SRN] · original work: S. Herbst, CS229 Stanford 2018

1. Problem statement

Neuroscience labs film fruit flies to measure behaviour. Existing tools are either too slow to keep up with the video (FlyTracker runs about $10\times$ slower than real time) or need a GPU (DeepLabCut, LEAP). Real-time analysis matters for closed-loop experiments and for long recordings. Herbst built a CPU-only pipeline that, for a courting male–female pair, reports each fly’s position, sex and $0\text{--}360^\circ$ heading, plus the male’s two wing angles. He reported 84 FPS on 1530×1530 frames. **Our goal was a straight reproduction:** implement exactly what the paper and poster describe, measure the same quantities (Tables 1–2, Figs. 8–10), and record every point where the sources leave a gap or disagree.

2. Dataset

The data is the author’s public set (github.com/sgherbst/cs229-project, MIT licence, about 2 GB): a 15-minute grayscale courtship video cropped to the 1530×1530 circular well, and five test clips (9–22 s). The author labelled **326 frames** as points in LabelMe: head, body point and abdomen for each fly (fh/fp/fa, mh/mp/ma), plus a wing base (mp2) and two wing tips (mw) for the male. The frames come in five sets: 100 sampled uniformly, 33 with the flies touching, 50 with flies on the wall, and 143 chosen for wing motion. All 326 loaded with consistent labels. Each stage builds its own examples from these frames (contours, pairs, flies or wings), so the example counts differ per stage.

3. Approach

Each frame passes through four learned stages. Each turns the image into a few features and hands them to a simple model.

FlyCount. A fixed dark threshold inside the well keeps only the fly bodies. Each contour’s area goes to a Gini decision tree with three classes: no fly, one fly, or both flies touching. $\sigma\text{♀}$ vs $\text{♀}\sigma$. For two one-fly contours, four features (area and moment-based aspect ratio of each) are standardized and a logistic regression decides whether the pair is ordered male–female. Each pair is also added in swapped order with the label flipped. **Orientation.** Central moments give the body axis, $\theta = \frac{1}{2} \text{atan}(2\mu_{11}/(\mu_{20}-\mu_{02}))$, but not which end is the head. The fly is rotated upright and cropped to 128×64 , and its HOG descriptor (8×8 cells, 9 bins, 3780-d) is reduced to 15 principal components. A logistic regression then decides whether to add 180° . There are separate male and female models, and every example is also added rotated 180° with the label inverted. **WingAngle.** Median blur, a threshold and erosion keep the male’s wings and remove the legs. The upright right half is cropped to 96×80 , described by HOG (3564-d), reduced with PCA-40, and fed to closed-form least squares $\theta = (X^T X)^{-1} X^T y$, which gives the angle from the body axis ($0\text{--}90^\circ$). The left wing is the same model applied to the mirrored image.

4. Implementation

The code is Python and uses only the libraries the paper names: scikit-learn, OpenCV, NumPy, Matplotlib, imutils, joblib and tqdm. Everything runs on the CPU. `src/` has one module per stage plus `train.py` (fits and saves all models, about 40 s), `evaluate.py` (Table 1), `profile.py` (Table 2), `figures.py` and `demo.py` (a live overlay on video). The paper leaves out many constants, among them the threshold of 115, the blur, erosion and HOG block sizes, the crop windows, the solver and how wing labels are computed. We took each one from the author’s public code, marked it *UNSPECIFIED* in our source, and listed it in NOTES.md. Training is deterministic (seed 0). OpenCV is pinned below version 5, which removed HOGDescriptor.

5. Results

Model	Train err. (ours / paper)	Test err. (ours / paper)	N train (ours / paper)	N test
FlyCount	0.0% / 0.0%	0.0% / 0.0%	759 / 759	253 / 253
$\sigma\text{♀}$ vs $\text{♀}\sigma$	0.0% / 0.2%	0.7% / 0.7%	423 / 423	141 / 141
σ orientation	0.0% / 0.0%	0.0% / 0.0%	358 / 358	120 / 120
♀ orientation	0.8% / 0.8%	0.0% / 0.0%	261 / 259	87 / 87
WingAngle	σ 2.35° / 2.06°	σ 2.18° / 2.92°	354 / 354	118 / 118

Table 1. Model accuracy, ours against the paper. WingAngle mean test error is 0.01° (paper 0.59°).

The three classifiers reproduce the paper’s test errors exactly. The example counts match to the unit except one extra female fly. The wing regression is slightly better on test, but with 118 test examples and one random split that difference is within noise. The poster’s ablation also holds: without the aspect-ratio features, $\sigma\text{♀}$ test error rises to 2.1% (poster 1.4%). Our FlyCount tree splits at 1112.5 and 10442.5. The second split matches Fig. 4 exactly; the first is a midpoint between training examples, so it moves with the split. Fig. 8 (two orientation PCs already separate upright from flipped flies) and Fig. 9 (wing angle almost linear in PC1) reproduce clearly. With 15 PCs, orientation explains 61% (σ) and 65% (♀) of the variance; the paper reports about 60%.

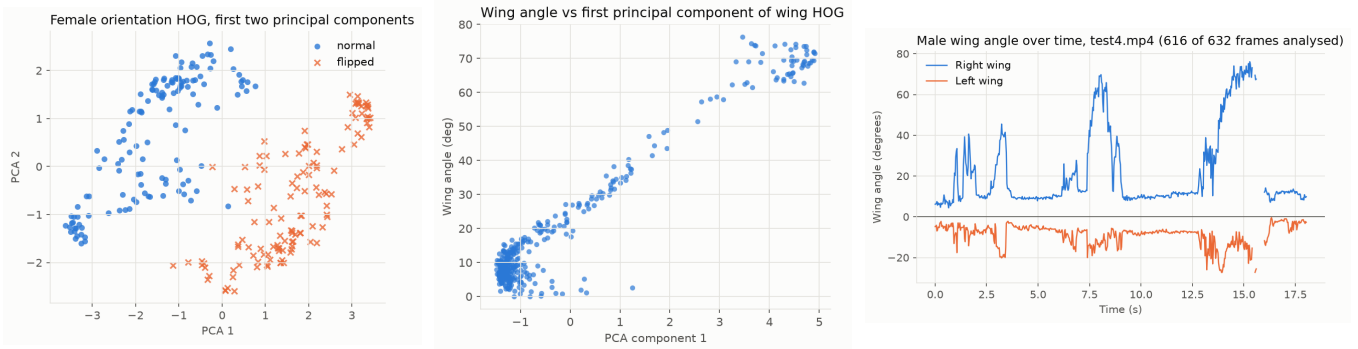


Figure 1. Reproduced Figs. 8, 9 and 10: female orientation HOG on PC1/PC2; wing angle vs PC1; male wing angles over test4.mp4 (left wing plotted negative). The right/left correlation is -0.69 (paper -0.64).

Step	I/O	FlyCount	σ^2 vs σ^2	Orientation	WingAngle	Throughput
Ours (ms)	4.3	4.5	0.8	1.7	2.5	72.5 FPS
Paper (ms)	3.9	2.2	0.7	1.9	3.3	84.0 FPS

Table 2. Per-frame timing. Ours: i7-13700HX, 16 GB RAM, no GPU, median of 5 passes over test4.mp4. Paper: 2.8 GHz Core i7, 16 GB RAM.

The pipeline runs at **72.5 FPS**, $2.4\times$ the 30 FPS source rate, so the real-time claim holds on different hardware. Frame reading is the largest single cost, as in the paper. Individual passes on this laptop ranged from about 22 to 74 FPS depending on background load, which is why we report the median.

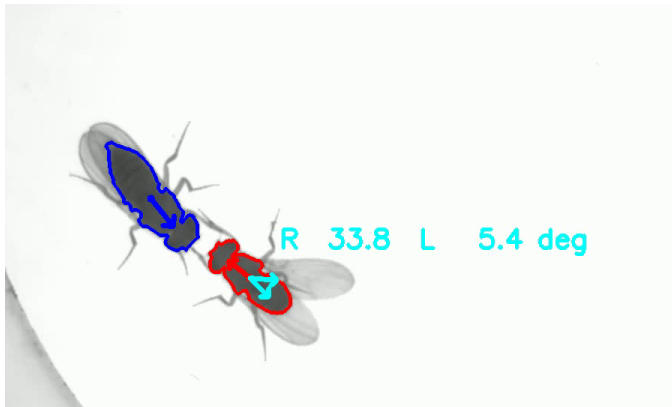


Figure 2. Live demo output on test4.mp4 (frame 300). The female (blue) and male (red) are outlined with heading arrows. Cyan arrows and text give the male's wing angles: right 33.8° , left 5.4° . Run it with python `src/demo.py`.

6. Conclusions

The paper's results reproduce. All Table 1 classifier errors match, the wing regression is within noise, the figures look the same, and the pipeline runs faster than real time without a GPU. As the author concludes, preprocessing does most of the work: thresholds, erosion and upright crops strip out everything except the variable of interest, so a HOG descriptor with 15–40 principal components and a linear model is enough. We found four discrepancies in the sources. (i) The text says one third is held out, but every Table 1 row is a 25% split, which we followed. (ii) Data is augmented before the split, so a swapped or rotated twin of a training example can land in the test set. This makes the test errors slightly optimistic, and we kept it to stay faithful. (iii) The "adaptive" threshold is a threshold set from each frame's mean. (iv) The claimed $2.4\times$ speed-up matches the clips' actual 35 FPS, not the poster's 30 FPS. Limitations remain: one video, one random split, and no way to separate flies that are touching.